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Industrial Policy Implementation: Empirical Evidence from China's

Shipbuilding Industry

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1 What this paper is doing

This paper studies *how* industrial policy is implemented in practice and what the *long-run* consequences are for market structure and performance, using China’s shipbuilding industry (1998–2014) as the laboratory. The authors build and estimate a dynamic empirical IO model with heterogeneous firms that make **entry**, **exit**, **investment**, and **Cournot production** choices in multiple ship product markets. The key challenge is that subsidies are not directly observed; therefore, the paper *infers* the magnitude of production, investment, and entry subsidies from observed outcomes and estimated cost structures before/after policy regimes. With the estimated model, the paper runs counterfactual simulations to compare industrial policy instruments (production vs. investment vs. entry subsidies), policy timing (pro-cyclical vs. counter-cyclical), and targeting rules (e.g. “White List” consolidation), focusing on discounted long-run industry revenue and net profit relative to subsidy costs.

2 Institutional background and empirical motivation

2.1 Industry and policy timeline

The shipbuilding market has large global demand swings (booms and busts), substantial fixed costs, and multi-year order backlogs. China implemented major policy interventions around:

- **2006–2008:** strong support (entry encouragement in some provinces; production and investment support), coinciding with a global boom;
- **2009+:** after the global financial crisis, policy emphasis shifts toward consolidation and support of selected incumbents; entry restrictions rise; investment support becomes stronger.

The empirical patterns in Figure 1 are the motivating facts: China’s market share rises sharply, entry accelerates, and investment jumps in the mid-2000s, while ship prices fall after 2008.

2.2 Figure 1 and Table 1: what the raw data say

Figure 1 provides four core time-series facts:

1. China becomes the largest producer: market share rises dramatically over the sample.
2. Entry is concentrated in the mid-2000s: a large wave of new yards appears before the post-crisis consolidation phase.
3. Investment by Chinese yards rises sharply around the policy/boom period.
4. Ship prices exhibit a boom-bust cycle: rising in the boom and collapsing after the crisis.

Table 1 summarizes key cross-sectional moments that discipline the structural model: average production, prices, backlog, capital, and the fraction of time yards are idle.

Table 1: Summary statistics

Panel A: Shipyards		
Variable	Mean	Std. dev.
Production (mCGT)	0.06	0.15
Price (RMB1,000/CGT)	12.11	5.50
Backlog	0.06	0.18
Capital (RMB billion)	0.53	0.75
Employment (1,000 workers)	1.68	2.37
Output per employee (mCGT/worker)	0.02	0.03
Steel price (RMB1,000/Ton)	3.35	1.09
Age (years)	12.70	14.44
Large firm indicator	0.12	0.32
Fraction of time idle	0.12	0.32
Panel B: Markets		
Quantity (mCGT)	23.71	12.76
Price (RMB1,000/CGT)	12.11	5.50
Earnings ratio (freight rates, 1,000 RMB)	0.19	0.15
Backlog (mCGT)	127.70	69.83
Steel price (RMB1,000/Ton)	3.35	1.09

Notes: Values are reported in the paper. “mCGT” denotes million compensated gross tonnage (a shipbuilding capacity/output metric).

3 The model: static Cournot competition embedded in firm dynamics

3.1 Players, products, and timing

Time is discrete (quarters). Each shipyard j can produce up to $M = 3$ ship types indexed by $m \in \{1, 2, 3\}$ (bulk carrier, tanker, containership). Each period has:

1. **Production stage:** given states, firms choose output q_{jmt} in each product market with Cournot competition.
2. **Exit stage:** after production, the firm draws an outside option/scrap value ϕ_{jt} and may exit.
3. **Investment stage:** conditional on staying, the firm draws an investment cost shock ν_{jt} and chooses investment $i_{jt} \geq 0$ to adjust future capital.
4. **Entry stage:** a pool of potential entrants draws an entry cost and decides whether to enter.

3.2 Ship demand

For each ship type m , industry inverse demand is written as

$$P_{mt} = P_m(Q_{mt}, d_{mt}), \quad Q_{mt} \equiv \sum_j q_{jmt}, \quad (1)$$

where d_{mt} is a vector of demand shifters (e.g. freight rates, backlog).

Linear demand for estimation. In the empirical implementation, demand is estimated in the (direct) form

$$Q_{mt} = \alpha_{0m} + \alpha_{pm}P_{mt} + d'_{mt}\alpha_{dm} + \varepsilon_{mt}. \quad (2)$$

This implies an inverse demand slope $\partial P_{mt}/\partial Q_{mt} = 1/\alpha_{pm}$.

3.3 Production costs and Cournot first-order condition

Let the variable cost for firm j in market m be $c_m(q_{jmt}, s_{jmt}, \omega_{jmt})$. Marginal cost is

$$MC_{jmt} \equiv \frac{\partial c_m(q_{jmt}, s_{jmt}, \omega_{jmt})}{\partial q_{jmt}},$$

and each operating firm also pays a fixed cost c_0 each period.

Cournot FOC (derivation). Firm j 's per-period variable profit from type m is

$$\pi_{jmt} = P_{mt}(Q_{mt}, d_{mt})q_{jmt} - c_m(q_{jmt}, s_{jmt}, \omega_{jmt}).$$

Under Cournot competition, firm j takes rivals' quantities as given, so $\partial Q_{mt}/\partial q_{jmt} = 1$. The first-order condition for an interior optimum is:

$$0 = \frac{\partial \pi_{jmt}}{\partial q_{jmt}} = P_{mt} + q_{jmt} \frac{\partial P_{mt}}{\partial Q_{mt}} - MC_{jmt}, \quad (3)$$

equivalently

$$P_{mt} + q_{jmt} \frac{\partial P_{mt}}{\partial Q_{mt}} = MC_{jmt}. \quad (4)$$

This is the paper's core static condition linking prices, quantities, and marginal costs.

Marginal cost specification. The empirical marginal cost function is specified as

$$MC_{jmt} = \beta_{0m} + s'_{jmt}\beta_{sm} + \beta_{qm}q_{jmt} + \omega_{jmt}, \quad (5)$$

where s_{jmt} contains observed cost shifters (steel price, capital, backlog, region and ownership indicators, policy indicators, etc.) and ω_{jmt} is an unobserved marginal-cost shock.

Closed-form optimal quantity under linear demand. Using $\partial P/\partial Q = 1/\alpha_{pm}$ from (2) and plugging (5) into (4), we get:

$$P_{mt} + \frac{1}{\alpha_{pm}}q_{jmt} = \beta_{0m} + s'_{jmt}\beta_{sm} + \beta_{qm}q_{jmt} + \omega_{jmt}, \quad (6)$$

$$\left(\beta_{qm} - \frac{1}{\alpha_{pm}}\right)q_{jmt} = P_{mt} - \beta_{0m} - s'_{jmt}\beta_{sm} - \omega_{jmt}. \quad (7)$$

Hence

$$q_{jmt}^* = \max \left\{ 0, \frac{1}{\beta_{qm} - 1/\alpha_{pm}} \left(P_{mt} - \beta_{0m} - s'_{jmt}\beta_{sm} - \omega_{jmt} \right) \right\}, \quad (8)$$

where the $\max\{\cdot, 0\}$ reflects the non-negativity constraint on output.

3.4 Dynamic problem: exit and investment

Let s_{jt} denote the payoff-relevant observed state variables for firm j at time t (capital, backlog/cost index, age, ownership, policy regime, and aggregate prices).

3.4.1 Exit and continuation value

After producing and earning static profits $\pi(s_{jt})$, the firm draws a scrap value ϕ_{jt} and chooses whether to exit:

$$V(s_{jt}, \phi_{jt}) = \pi(s_{jt}) + \max\{\phi_{jt}, CV(s_{jt})\}. \quad (9)$$

The continuation value is

$$CV(s_{jt}) = \mathbb{E}_\nu [-C_i(i^*(s_{jt}, \nu_{jt}), s_{jt}, \nu_{jt}) + \beta \mathbb{E}[V(s_{j,t+1}) \mid s_{jt}, i^*(s_{jt}, \nu_{jt})]]. \quad (10)$$

Exit probability. Exit occurs when $\phi_{jt} > CV(s_{jt})$. Therefore

$$p_x(s_{jt}) \equiv \mathbb{P}(\phi_{jt} > CV(s_{jt})) = 1 - F_\phi(CV(s_{jt})), \quad (11)$$

where F_ϕ is the CDF of ϕ .

3.4.2 Investment

Capital evolves as

$$k_{j,t+1} = (1 - \delta)k_{jt} + i_{jt}.$$

The paper assumes a convex investment cost with an unobserved cost shifter:

$$C_i(i_{jt}, s_{jt}, \nu_{jt}) = c_1(s_{jt}) i_{jt} + c_2 i_{jt}^2 + c_3 \nu_{jt} i_{jt}. \quad (12)$$

The optimal investment $i^*(s_{jt}, \nu_{jt}) \geq 0$ satisfies the Kuhn–Tucker condition

$$\beta \frac{\partial \mathbb{E}[V(s_{j,t+1}) \mid s_{jt}, i]}{\partial i} \Big|_{i=i^*} \leq \frac{\partial C_i(i, s_{jt}, \nu_{jt})}{\partial i} \Big|_{i=i^*}, \quad \text{with equality if } i^* > 0. \quad (13)$$

Given (12), the marginal investment cost is

$$\frac{\partial C_i}{\partial i} = c_1(s_{jt}) + 2c_2 i_{jt} + c_3 \nu_{jt}.$$

Industrial policy affects investment by shifting $c_1(s_{jt})$ (interpretable as an investment subsidy rate).

3.5 Entry

Each period, $\bar{N}$ potential entrants draw an entry cost κ_{jt} and decide whether to enter. If they enter, they pay κ_{jt} and become incumbents next period with an initial capital draw. The entry decision is summarized by an entry cutoff: enter if $\kappa_{jt} \leq V^E(s_t)$, where V^E is the value of entry (net of entry cost). Entry probability is therefore

$$p_e(s_t) \equiv \mathbb{P}(\kappa_{jt} \leq V^E(s_t)) = F_\kappa(V^E(s_t) \mid s_t), \quad (14)$$

with F_κ shifted by policy regimes (entry subsidies or entry restrictions).

4 Estimation strategy (what is identified from what)

The paper uses a two-stage approach (Hotz–Miller / Bajari–Benkard–Levin style): first estimate static primitives and policy functions, then recover dynamic parameters using an approximated value function.

4.1 Stage 1: demand and production costs

- **Demand:** estimate (2) using market-level data with instruments for price (details in the paper’s Supplementary Material).
- **Marginal costs:** use (8) implied by Cournot and linear demand to estimate $\{\beta_{0m}, \beta_{sm}, \beta_{qm}, \sigma_{\omega m}\}$ via a Tobit likelihood (because $q_{jmt} \geq 0$ and many observations are at zero).
- **Fixed costs:** calibrate c_0 from accounting data (important because idling is common).

4.2 Stage 2: dynamic parameters (scrap value, investment costs, entry costs)

- **Exit policy:** estimate $p_x(s)$ flexibly (probit on a polynomial in states).
- **Investment policy:** recover $i^*(s, \nu)$ by exploiting monotonicity in ν and the relationship between investment quantiles and ν quantiles; impose an additive structure

$$i^*(s, \nu) = h_1(s) + h_2(\nu),$$

and normalize $\nu \sim N(0, 1)$ to pin down scales.

- **State transitions:** model ship prices and steel prices as (regime-dependent) AR(1) processes; update capital deterministically using depreciation and investment.
- **Value function approximation:** approximate the ex-ante value $V(s)$ using B-spline basis functions.

Key simplification under exponential scrap value. If $\phi \sim \text{Exponential}(\text{mean } \sigma_\phi)$, then

$$\mathbb{E}(\phi \mid \phi > CV) = CV + \sigma_\phi.$$

Using $p_x(s) = \mathbb{P}(\phi > CV(s))$, the ex-ante value satisfies

$$V(s) = \pi(s) + CV(s) + p_x(s)\sigma_\phi. \tag{15}$$

This identity is central because it makes the value function easier to compute once $p_x(\cdot)$ and $CV(\cdot)$ are available.

5 Main parameter estimates and what they mean

5.1 Table 2: cost function estimates (production subsidy and cost shifters)

Table 2 reports marginal cost estimates for Chinese shipyards.

Table 2: Cost function estimates

	Bulk carrier		Tanker		Containership	
	Coeff.	T-stat	Coeff.	T-stat	Coeff.	T-stat
Type-specific						
β_q	7.29	7.59	14.13	5.10	10.58	5.01
σ_ω	9.58	8.93	16.27	6.91	13.77	5.14
Constant (1,000 RMB/CGT)	20.37	14.05	39.71	8.78	34.92	7.27
Steel Price (1,000 RMB/Ton)	1.68	6.85	1.14	2.83	0.66	1.50
Capital (Bn RMB)	-2.67	-2.85	-2.89	-1.74	-2.44	-1.93
Capital ²	0.20	0.80	0.07	0.24	0.06	0.28
Backlog	-1.80	-5.03	-5.02	-4.97	-3.30	-3.19
Backlog ²	0.08	3.94	0.26	3.44	0.20	1.94
Backlog of other types	0.13	0.86	0.38	1.57	0.53	2.61
Common (reported once in the paper table; applied across types)						
2006–8	-2.10	-3.01	–	–	–	–
2009+	-1.22	-1.78	–	–	–	–
Large firms	-4.32	-6.54	–	–	–	–
Jiangsu	-2.96	-4.61	–	–	–	–
Zhejiang	-1.62	-2.80	–	–	–	–
Liaoning	-2.10	-2.01	–	–	–	–
CSSC/CSIC	-0.86	-1.17	–	–	–	–
Private	0.16	0.30	–	–	–	–
Foreign JV	-0.86	-1.41	–	–	–	–
Age	0.21	3.22	–	–	–	–
N	4,886		4,977		2,504	

Notes (interpretation): coefficients are for marginal cost in 1,000 RMB per CGT. The policy dummies (2006–8, 2009+) are interpreted as *production subsidies* that lower marginal cost. “Common” coefficients are constrained to be the same across ship types but are displayed once in the paper’s table.

Interpretation (key takeaways).

- **Production subsidy magnitude:** the 2006–2008 dummy is about -2.10 (thousand RMB/CGT), implying a sizeable marginal-cost reduction in that period; the post-2009 subsidy is smaller (-1.22).
- **Convexity and congestion:** positive β_q means marginal costs rise with own output (increasing marginal cost). Backlog has a negative linear and positive quadratic term: moderate backlog can reduce marginal cost (scale/learning/organizational efficiency), but too much backlog increases costs (capacity strain).
- **Capital lowers marginal cost:** negative capital coefficients indicate capacity investment reduces marginal costs; the small positive capital-squared term suggests diminishing returns.
- **Heterogeneity:** large firms and some provinces (e.g. Jiangsu, Zhejiang) have lower marginal costs, consistent with systematic productivity differences.

5.2 Table 3: investment cost and scrap value parameters (investment subsidy channel)

Table 3 estimates the investment-cost curvature and the mean scrap value parameter.

Table 3: Estimates of investment cost and scrap value parameters

	Coeff.	T-stat
σ_ϕ	0.98	12.32
$c_1 \cdot \mathbf{1}\{2006-8\}$	-0.27	-1.70
$c_1 \cdot \mathbf{1}\{2009+\}$	-0.46	-3.27
c_2	29.54	14.49
c_3	2.07	9.67
N	4,286	

Notes (interpretation): investment and scrap values are in billions of 2000 RMB. Negative c_1 shifts are interpreted as proportional investment subsidies that reduce the marginal (linear) investment cost in those policy regimes.

Interpretation.

- **Scrap value scale:** $\sigma_\phi \approx 0.98$ (bn RMB) pins down the exit option distribution and thus exit sensitivity to continuation values.
- **Investment subsidy strengthens after 2009:** the post-2009 shift (-0.46) is larger in magnitude than the 2006–8 shift (-0.27), rationalizing why investment remains high even when ship prices fall.
- **Convex adjustment costs:** large c_2 implies strong convexity; marginal investment costs rise quickly with investment, so subsidizing during booms can induce expensive overexpansion.
- **Unobserved investment wedges:** c_3 and the shock ν capture substantial heterogeneity in investment behaviour among observationally similar firms.

5.3 Table 4: entry cost distribution by province (entry subsidy / restriction channel)

Table 4: Entry cost distribution mean by province (bn 2000 RMB)

	κ_{pre}	$\kappa_{\text{post},06}$	% of pre	$\kappa_{\text{post},09+}$	% of pre
Jiangsu	86	31	36	91	106
Zhejiang	133	54	41	264	199
Liaoning	82	40	49	–	–
Other	38	15	38	61	160

Notes (interpretation): lower κ means lower effective entry cost (consistent with entry subsidies or easier entry). Higher post-2009 κ indicates entry became much more difficult (consistent with consolidation/moratorium).

Interpretation. Entry costs fall sharply in the “entry encouragement” regime and then rise substantially in the post-2009 regime, consistent with a policy shift from expansion to consolidation. The geographic heterogeneity (e.g. Zhejiang) reflects province-level policy differences.

6 Counterfactual policy evaluation: what works, what fails, and why

6.1 Figures 2–3: the effect of having subsidies at all

- **Investment (Figure 2):** with subsidies, total investment is much higher; without subsidies, investment is dramatically lower during 2006–2013. This matches the idea that production/investment subsidies increase expected continuation values and reduce effective investment costs.
- **Number of firms (Figure 3):** subsidies raise entry and reduce exit, leading to a larger number of operating yards and more fragmentation (lower concentration and lower utilization).

6.2 Table 5: Comparing Production vs. Investment vs. Entry Subsidies

Table 5 is the paper’s central “instrument comparison” table.

Table 5: Comparison of policy instruments (bn 2000 RMB, discounted 2006–2050)

	All subsidies	Production subsidy	Investment subsidy	Entry subsidy	Remove all
Revenue	2361	2154	1873	1961	1810
Net Profit	1085	1061	981	1023	950
Prod. Subsidy	262	225	0	0	0
Inv. Subsidy	77	0	42	0	0
Entry Subsidy	431	0	0	231	0
Δ Rev./Subsidy (%)	72	153	153	66	–
Δ Profit/Subsidy (%)	18	50	74	32	–

Notes: “Net Profit” = revenue – production/investment costs + scrap value – entry costs. Last two rows show effectiveness: incremental revenue/profit per subsidy spending.

Interpretation (why entry subsidies are wasteful).

- Entry subsidies increase the number of firms directly, but they tend to attract smaller, higher-cost entrants. That raises fragmentation and idle capacity, and delivers low incremental profit per RMB.
- Production and investment subsidies are more effective in raising revenue and profit, because they (i) apply to incumbents with existing scale/capacity and (ii) amplify scale economies via backlog and capital accumulation.
- The paper emphasizes **convex distortions**: combining several subsidies is less effective than using one in isolation, because the policy mix inflates the operating mass of firms and makes competition and underutilization more severe.

6.3 Table 6 and Figure 4: why counter-cyclical policy dominates pro-cyclical policy

The paper compares subsidizing during the boom (2006–2008) vs. subsidizing during the bust (2009–2013), holding subsidy spending comparable.

Table 6: Pro-cyclical vs. counter-cyclical policy (bn 2000 RMB)

	Subsidize during boom	Subsidize during recession
Lifetime Revenue 2006–	1880	1872
Lifetime Profits 2006–	961	975
Production subsidy	29	29
Investment subsidy	13	14
Δ Revenue / Subsidy (%)	189	168
Δ Net profit / Subsidy (%)	38	70

Notes (interpretation): No entry subsidies in these scenarios. Subsidy rates are adjusted so that spending is comparable across the two experiments.

Interpretation. The key result is that subsidizing during the downturn yields a much larger return in net-profit terms (70% vs. 38%). The mechanisms are:

- **Convexity:** marginal and adjustment costs are higher when the industry is already near capacity in the boom, so additional subsidized expansion is expensive (Figure 4’s marginal-cost index is higher).
- **Composition:** booms attract marginal, inefficient entrants; busts keep relatively efficient incumbents alive and mobilize idle capacity.

6.4 Table 7: Permanent vs. Temporary vs. Targeted Temporary Policies

The paper also varies duration and targeting.

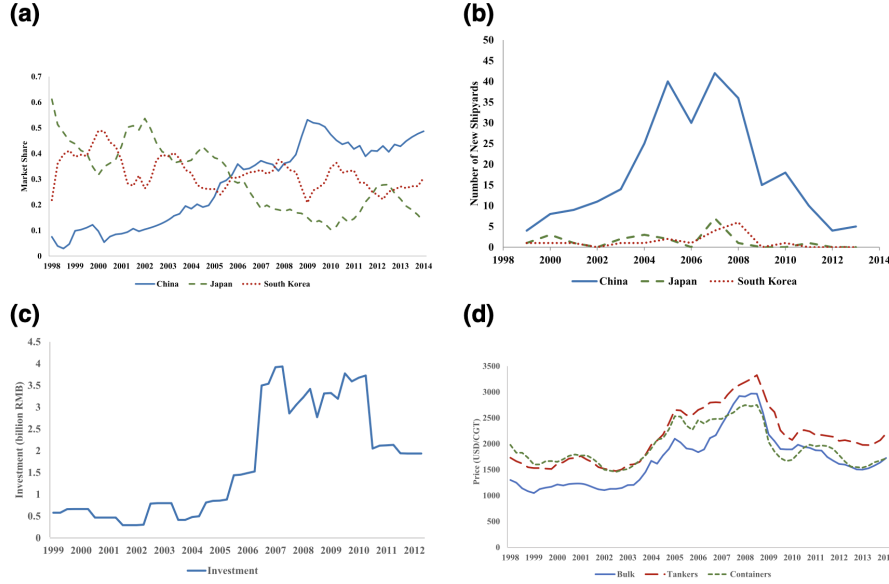


FIGURE 1

Table 7: Alternative policy designs (bn 2000 RMB, discounted 2006–2050)

	Permanent subsidy	Temporary (2006–13)	Targeted Temp. (2006–13)	No subsidies
Revenue	1708	1655	1614	1366
Net Profit	969	991	1035	853
Prod. Subsidy	193	201	191	0
Inv. Subsidy	55	49	54	0
Entry Subsidy	0	0	0	0
Total Subsidy	248	250	245	0
Δ Rev./Subsidy (%)	138	116	101	—
Δ Profit/Subsidy (%)	47	55	75	—

Notes: Temporary policies avoid long-run selection distortions; targeting further raises returns by shifting support toward efficient firms.

Interpretation. Temporary subsidies improve the rate of return (47% $\rightarrow$ 55%) by reducing long-run entry and protecting fewer inefficient firms (a *selection* channel). Adding targeting improves returns further (to 75%) by reallocating support toward low-cost/high-profit firms (a *misallocation reduction* channel). These results connect to a broad IO/macro theme: policy design affects not just levels (output) but the *composition* of producers and hence long-run productivity and capacity utilization.

7 Short summary

1. **Industrial policy implementation can be evaluated instrument-by-instrument.** Output/production and investment subsidies can raise long-run revenue, but the return depends on timing and targeting.
2. **Entry subsidies are the most distortionary margin in this setting.** They inflate the mass of high-cost firms and lead to fragmentation and idle capacity.

3. **Convex costs and business cycles matter for policy design.** Counter-cyclical support can dominate pro-cyclical support because expansion is cheaper in downturns and composition is better.
4. **Targeting and temporariness increase effectiveness.** They mitigate selection distortions and concentrate support where it generates larger long-run benefits.

8 Conclusion

8.1 Summary

The paper’s central conclusion is that China’s shipbuilding industrial policy *worked* in the narrow sense that it rapidly expanded domestic capacity and raised China’s world market share, but it also generated substantial *dynamic and compositional distortions*. Using firm-level data (1998–2014) and an estimated dynamic model with heterogeneous firms choosing entry, exit, investment, and Cournot production, the authors infer large subsidies that are not directly observed in the data. Counterfactual simulations show that subsidies boosted investment and entry and contributed to lower world ship prices, but they also intensified industry fragmentation and increased idle capacity, which implies a low welfare return on subsidy spending in the early policy phase. Effectiveness improves when policy moves away from “support everyone” toward more selective support of efficient producers. The general welfare lesson is therefore about *policy design*: counter-cyclical intervention, targeting high-productivity firms, and limiting the duration of support can substantially reduce distortions while preserving the intended industrial expansion.

8.2 Main findings to remember (from the model + counterfactuals)

1. **Big aggregate effects, but low early returns.** Subsidies can explain the observed surge in China’s investment, entry, and global market share, yet the implied rate of return on subsidy costs is initially low because the policy expands capacity in a way that depresses prices and leaves many yards underutilized.
2. **Instrument ranking: production/investment vs. entry.** Production and investment subsidies can be rationalized if the objective is market share or strategic capacity building, but entry subsidies are comparatively wasteful because they attract many marginal producers and worsen long-run fragmentation (see the instrument comparison results summarized in Table 5 of the paper).
3. **Timing matters: counter-cyclical support dominates pro-cyclical support.** Because shipbuilding features large boom–bust cycles and convex costs, subsidizing during downturns yields higher net-profit gains per unit of subsidy than subsidizing during booms (Table 6).
4. **Targeting and temporariness raise welfare.** Subsidies are more effective when they are temporary rather than permanent, and when they are targeted toward efficient firms rather than spread across all firms (Table 7).
5. **Rationales: beyond the classical textbook story.** The evidence is more consistent with nonclassical motives—such as lowering freight rates to support trade—than with some standard rationales like industry-wide knowledge spillovers; there is also suggestive evidence consistent with military considerations (Section 7 and the paper’s Supplementary Material).

8.3 Economic intuition: why these results arise (one continuous mechanism)

The model's mechanism can be summarized as a single causal chain:

1. **Subsidies shift dynamic payoffs and therefore firm states.** Production subsidies reduce marginal cost today; investment subsidies reduce the marginal cost of building capacity; entry subsidies reduce the cost of becoming an incumbent. All three raise the continuation value and thus stimulate participation and expansion.
2. **But the extensive margin is the most distortionary margin in this industry.** When entry is subsidized, many relatively high-cost/small yards enter. In a Cournot setting, additional firms intensify competition and push down prices. With fixed operating costs and demand volatility, the industry then displays *fragmentation* and *idle capacity*: too many yards chase too little demand in bust periods.
3. **Exit is sluggish, so mistakes persist.** Because firms face option value and heterogeneous scrap values, low demand does not immediately clear the market through exit. Instead, many firms remain and stay idle, creating persistent overcapacity (a hysteresis-like pattern).
4. **Convex investment costs make pro-cyclical expansion expensive.** With convex adjustment costs, expanding when demand is already high (booms) has a high marginal resource cost. Counter-cyclical subsidies exploit lower marginal costs in downturns and keep efficient capacity alive, producing higher net benefits per unit of subsidy.
5. **Heterogeneity creates a composition channel, so targeting helps.** Firms differ in cost and scale. Untargeted subsidies keep inefficient yards alive and encourage inefficient entry, worsening allocation. Targeted subsidies shift output and survival toward low-cost firms, improving the industry composition and raising net profits relative to subsidy spending.

8.4 Takeaway

Industrial policy can rapidly build market share, but in industries with volatile demand, fixed costs, and heterogeneous firms, “who enters and survives” is the key margin: counter-cyclical, temporary, and targeted support dominates broad, pro-cyclical entry promotion.